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Register Number :

HALF YEARLY EXAMINATION - 2022

BIOLOGY

Std: 12

Marks : 70
Time : 3.00 hr

PART I - (Bio-Botany) (Marks – 35)

PART – I

I. Answer all the questions 8x1=8

Choose the most appropriate answer from the given four alternatives and write the option and the corresponding answer.

1. Identify the correctly matched pair.
   a. Tuber - Allium cepa
   b. Sucker - Pistia
   c. Rhizome - Musa
   d. Stolen - Zingiber.

P. Sanjeev raj
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2. Which of the following is not a biotic component of the ecosystem.
   a. Organic Components b. Bacteria c. Inorganic compounds d. Humus
3. Match the following
   A. A pair of chromosomes Extra with diploid - i. Monosomy
   B. one chromosomes extra To the diploid - ii. tretrasomy
   C. One chromosome loses From diploid - iii. Trisomy
   D. Two individual chromosomes lose from diploid - iv. Double monosomy
4. In which techniques Ethidium Bormide is used
   a. polymerized chain Reaction b. Southern Blotting Techniques
   c. Agarose Gel Electrophoresis d. Western Blotting Techniques
5. In the following plants which one is called as 'king of bitters'
   a. Nilavembu b. keezhanelli c. vembu d. Pepper
6. . . . . is a modification of a biological structure where by an ancestral frait reappears after having been lost through reemergence of sexual reproduction.
   a. Atavism b. Receive organism c. hybrids d. Dominant organism
7. The selected tissue to establish a new Plant in a culture medium
   a. Embryoid b. callus c. platlets d. Explant
8. The unit for measuring ozones thickness a. Joule b. Kilos c. Dobson d. watt

PART –II

II. Answer any Four of the following 4x2=8

9. How will you identify a vector?
10. What is endothecium?
11. What is Test cross?
12. What are jumping genes?
13. What is Myromecophily?
14. Define – Cybrid.

PART – III

III. Answer any Three questions. Question No. 19 is compulsory 3x3=9

15. List out the functions of tapetum?
16. Distinguish habitat and niche
17. Draw and lable the structure of Embryo sac.
18. What is – Pollen kitt
19. Draw labeled structure of Plasmid $P^{BR}322$

PART – IV

Kindly send me your questions and answerkeys to us : Padasalai.net@gmail.com

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IV. Answer all the questions **2x5=10**

20. a. What is parthenocarphy? Write its significance (or)
    b. Explain in complete dominance with an example.
21. a. Write the Applications of plant tissue culture. (or)
    b. Shape of pyramid in a particular ecosystem is always different in shape, Explain with example.

BIO – ZOOLOGY (Marks : 35)

PART – I

I. Choose The correct Answerr **8x1=8**

1. The harmone inhibin is secreted by
   a. Leydig cells b. sertoli cells c. Skene's gland d. prostate gland
2. Albinism is caused due to the lack of the enzyme
   a. Thyroxine b. Thymulin c. Tyrosinase d. Amylase
3. A mRNA molecule is produced by
   a. Transcription b. Replication c. Duplication d. Tranlation
4. Who published the book 'Philosophie Zoologique'
   a. Darwin b. Spence c. Osborn d. Lamarck
5. The duration of Erythro cytic cycle in quartan Malaria is
   a. 48 hours b. 36 hours c. 72 hours d. 16 hours
6. Which of the following is referred to as industrial alcohol
   a. Methanol b. Ethanol c. Ethane d. Methane
7. The rule explains the relationship between lower the temperature and more the vertebrae in fishes a. Bergmann's rule b. Allen's rule c. Rench's rule d. Jordan's rule
8. Which of the following forest is known as the lungs of the planet earth?
   a. Tundra forest b. Taiga forest
   c. Amazon rain forest d. Rain forest of north east India

II. Answer any four of the Following **4x2=8**

9. Differentiate between the seasonal breeders and continuous breeders with example
10. What are Multiple Alleles?
11. What is meant by genophore?
12. What are Haptens?
13. What is bioremediation?
14. Write short notes on Ecosan toilets?

III. Answer Any three of the Following **3x3=9**

15. What is Aminocentesis? It is banned in our country, Is it necessary? Comment.
16. What are the salient features of Mutation theory?
17. Differentiate between somatic cell gene therapy and germ line gene therapy
18. Write a note on. i. Hibernation ii. Aestivation iii. Diapause
19. Tabulate the difference between Insitu and Exsitu conservation.

IV. Answer all the Questions **2x5=10**

20. a. Explain the various phases of the menstrual cycle. (Or)
    b. Why the human genome project is called a mega project? mention the main goals of human genome project.
21. a. Draw and explain the structure of HIV virus. (Or)
    b. Explain the three steps involved in sewage treatment.

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